

Model review and revisions

JMP progress, September 17–October 2, 2026

Tommaso De Santo

I reviewed the model and calibration, including preferences, credit, demographic accounting, and the empirical targets.

- Returned to a Stone–Geary housing requirement that applies only to parents, retaining nonlinear benefits from children.
- Tested equivalence scales and changing housing shares.
- Rebuilt the first-birth housing evidence and national input measures.
- Revised adult entry and tested how purchase timing affects the calibration.

Household decisions

Overlapping generations of households differ in age, earnings, wealth, housing tenure, and family size.

- During fertile ages, households choose whether to attempt a birth; success becomes less likely with age.
- They choose consumption, saving, housing size, and whether to rent or own; renters pay rent, while owners make a down payment and face selling costs.
- Persistent earnings risk and borrowing limits shape saving; a first-birth fixed cost and taste shocks shape fertility.
- Households retire, face mortality risk, and value bequests.

Children and housing needs

The first child requires basic space at home; each additional child raises the goods needs of a larger family.

$$u(c, s; m) = \frac{\left[\frac{c^{\alpha_0} s^{1-\alpha_0}}{e(m)} \right]^{1-\sigma}}{1-\sigma} + \psi m^{1-\gamma},$$
$$e(m) = \left(\frac{2 + 0.7m}{2} \right)^{0.7}, \quad s = \eta_\tau [h - h_P \mathbf{1}\{m > 0\}],$$
$$\eta_R = 1, \quad \eta_O = \chi.$$

Here h is rooms, m is children at home, and $\tau \in \{R, O\}$ denotes tenure. Parents require $h > h_P$; χ raises housing services for owners. The equivalence scale $e(m)$ increases family consumption needs. Child benefits are zero at $m = 0$; ψ sets their level and γ controls diminishing marginal benefits.

Demographic and market equilibrium

Children leaving home and new adult households entering the economy are separate transitions.

- Children leave dependency independently; adult entry follows earlier births with a lag.
- Positive estates finance entrant assets; any remainder leaves the economy.
- Stationary calibration normalizes population to one: the house price satisfies birth renewal, and housing clearing determines H_0 .
- A balanced payroll tax finances equal retiree pensions.

Housing around the first birth

I found a mistake in the earlier regression and rebuilt the PSID event study using interview dates and household status before birth.



At years 3–4, rooms rise by **1.465** (SE 0.050), ownership by **26.886 pp**, and recent moving falls by **21.147 pp**.

Reference person or spouse before birth. Baseline: years $-3/-2$; person and year effects; survey weights; 95% pointwise intervals.

Externally set parameters

Parameter	Value	Source
Model period	4 years	
Entry and retirement ages	18, 66	
Risk aversion σ	2	
Equivalence scale (ω, ν)	(0.7, 0.7)	Scholz et al. (2006)
Consumption share α_0	0.733	CEX 2019–23, childless renters
Annual real interest rate	2%	Greaney et al. (2025)
Annual depreciation; property tax	1.416%; 1.060%	National housing inputs
Financed share ϕ	0.80	20% down payment
Selling cost ψ_s	6%	Greaney et al. (2025)
Owner house sizes	2, 4, 6, 8, 10 rooms	
Largest rental unit	6 rooms	Greaney et al. (2025)
Housing-supply elasticity	0.63	Baum-Snow and Han (2024)
Pension / working-household earnings	0.229	CPS ASEC, income 2007
Payroll tax	8.0%	Balances the pension budget
Bequest wealth shift θ_1	0.008	1% of median annual earnings

Parameters and target moments

2007 stationary economy: ten parameters, eleven moments.

Parameter	Interpretation	Main target moment	Target
<i>Saving and bequests</i>			
β	Patience	Wealth / earnings	6.927
θ_0	Bequest weight	Annual child bequests / wealth (%)*	0.729
<i>Housing and tenure</i>			
χ	Owner housing-service premium	Ownership, ages 30–55 (%)	67.626
κ_T	Tenure-taste dispersion	Recent-parent ownership difference (pp)	12.761
h_P	Basic space for parents	First-birth rooms difference*	1.465
–	Additional moment	Mean rooms*	5.729
<i>Fertility</i>			
ψ	Child-benefit level	Completed fertility	2.100
ξ	First-birth cost	Childlessness, ages 40–44 (%)	19.828
κ_1	First-birth taste dispersion	Mean first-birth age	25.976
γ	Child-benefit curvature	One child among mothers (%)	21.366
κ_C	Later-birth taste dispersion	Children at age 25	0.810

Each parameter is paired with the moment it mainly moves; all ten are estimated jointly. *Recomputed: mean rooms (AHS), bequests (SCF), first-birth rooms (PSID).

Credit timing

A renter with financial wealth b and period income y who buys a home worth $Q = pH$:

$$c + (\delta + \tau_H) Q + b' = R(b - Q) + y.$$

Normally the buyer must have the 20% down payment at closing: $b \geq (1 - \phi)Q$, with $\phi = 0.8$.

The problem. A model period is four years. Requiring the down payment from wealth at the start of the period ignores saving within the period; counting all of it lets a buyer close with little equity and reach 20% only over the next four years. The rules differ in how much of the period's net saving $S = b' - (b - Q)$ counts:

$$b + \lambda S \geq (1 - \phi) Q.$$

- Soft, $\lambda = 1$: all of the period's saving counts.
- Quarter-saving, $\lambda = \frac{1}{4}$: a quarter of it counts, roughly one year of saving.
- Hard, $\lambda = 0$: only wealth brought into the period counts.

Other issues

- **Entrant wealth and income.** Revisited the wealth–income mapping after the earnings revision. Some negative-wealth entrants were infeasible without borrowing.
- **Empirical inputs.** Made housing, wealth and pension measures consistent in geography, units and income definitions.
- **Demographic accounting.** Checked the separate timing of children leaving home and adult entry, and how estates finance entrant assets.

Stone–Geary calibration

2007 stationary economy; 80% mortgage financing

Moment	Target	Soft	Hard	Quarter-saving
Completed fertility	2.100	2.100	2.100	2.100
Childlessness, ages 40–44 (%)	19.828	20.222	20.413	20.546
One child among mothers, ages 40–44 (%)	21.366	21.264	20.907	20.844
Mean first-birth age (years)	25.976	26.032	26.007	26.057
Wealth / earnings	6.927	6.551	6.610	6.645
Annual child bequests / wealth (%)	0.729	0.685	0.684	0.685
Mean occupied rooms	5.729	5.957	6.241	6.131
Ownership, ages 30–55 (%)	67.626	66.527	60.878	64.572
First-birth rooms difference	1.465	1.324	1.063	1.167
Recent-parent ownership difference (pp)	12.761	11.825	10.125	11.683
Children at age 25 (capped at three)	0.810	0.528	0.524	0.521
Validation moments	Target	Soft	Hard	Quarter-saving
First births at age 30+ (%)	24.928	23.448	23.078	23.395
Older wealth/income p90/p50	3.516	2.968	2.861	2.937
Family rooms difference	0.385	0.304	0.321	0.316

Parameter estimates

Estimates under the three purchase rules; H_0 follows from housing clearing.

Parameter	Economic interpretation	Soft	Hard	Quarter-saving
β	Annual discount factor	0.967	0.967	0.968
χ	Owner housing-service premium	1.097	1.068	1.091
ξ	First-birth fixed utility cost	0.352	0.385	0.397
κ_1	First-birth taste dispersion	0.124	0.123	0.128
κ_C	Later-birth taste dispersion	0.363	0.350	0.348
θ_0	Bequest weight	0.106	0.106	0.109
γ	Child-benefit curvature	0.101	0.101	0.098
κ_T	Tenure-taste dispersion	0.014	0.014	0.015
ψ	Child-benefit level	0.178	0.172	0.174
h_P	Basic space requirement for parents (rooms)	2.504	2.562	2.492
H_0	Housing supply coefficient	6.757	7.546	7.289

Credit and the long-run equilibrium

Permanent increase in mortgage financing from 80% to 100%

Keep preferences and housing supply scale H_0 fixed. Recompute house prices and population; each initial equilibrium has $N = 1$.

	Hard		Quarter-saving	
	80%	100%	80%	100%
Population N	1.0000	0.9768	1.0000	0.9789
House price	0.6566	0.6533	0.6745	0.6684
Ownership, all ages (%)	60.53	69.13	64.05	73.11
Completed fertility	2.100	2.100	2.100	2.100

Why the response is small: what I have checked

Credit raises births only if a childless renter delays a child because, after a birth, renting would leave the family short of space and the down payment blocks buying. Three things must hold:

1. **After a birth, renters are short of space.** Yes: about 80% of new parents who keep renting are at the six-room cap.
2. **Being short of space is costly.** Not very: one room short costs about 2% of consumption for one period under Cobb–Douglas preferences.
3. **Credit makes the space cheaper.** No: a room costs the same per period rented or owned. Credit gives access to homes above six rooms, not cheaper space, and the ownership premium χ is the same for parents and non-parents.

So credit removes an obstacle that exists but might be small. I have not yet shown how much of the small response this accounts for.

Policies that may do more (not yet run):

- Property-tax cut: makes space cheaper for renters and owners.
- Credit only for parents: raises the value of having a child.

What I want to try next

- Selling costs: check whether the cost of trading up after a birth is what keeps credit from reaching fertility, and whether the 6% seller-paid cost is the right convention.
- Small owner homes: young buyers may lock into homes too small for a family.
- A housing requirement that grows with the number of children.
- Improve the calibration.
- Reconsider the utility specification.